

ASSESSMENT FORM

Course: COMP6839031- Program Design Methods

Method of Assessment: Case Study

Semester/Academic Year : 1/2024-2025

Name of Lecturer : Team
Date : 09 September 2024 – 04 Januari 2025
Class : All Class Computer Science @Bandung
Topic : Input-Process-Output, Analysis Modeling, Business Process and Functional Modeling

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Student Outcomes:

SO 2 - Mampu merancang, mengimplementasikan, dan mengevaluasi solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi dalam konteks ilmu computer

Able to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of computer science.

SO 6 - Mampu untuk menerapkan teori ilmu komputer dan dasar-dasar pengembangan perangkat lunak untuk menghasilkan solusi berbasis komputasi.

Able to apply computer science theory and software development fundamentals to produce computing-based solutions

Learning Objectives:

(LObj 2.1) Mampu merancang solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi tertentu dalam konteks ilmu komputer

Able to design a computing-based solution to meet a given set of computing requirements in the context of computer science.

(LObj 6.1) Mampu menerapkan teori ilmu komputer yang digunakan untuk menghasilkan solusi berbasis komputasi

Able to apply computer science theory used to produce computing-based solutions

(LObj 6.2) Mampu menerapkan dasar-dasar pengembangan perangkat lunak untuk menghasilkan solusi berbasis komputasi

Able to apply software development fundamentals to produce computing-based solutions

Learning Outcomes:

LO 1: Explain the requirements of writing a program or system

LO 2: Apply pseudo-code to solve problem

LO 3: Analyze the requirements of a system

LO 4: Solve system design problems using UML

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
1	LO 1 LO 2 LObj 2.1 SO 2	Ability to identify requirements for programs	25%	Able to identify the input, process, and output of a program completely and correctly	Able to identify the input, process, and output of a program in less-completely and less-correctly	Able to identify the input, process, and output of a program in half-completely and half-correctly	Identify the input, process, and output of a program incompletely and incorrectly		
2	LO 2 LObj 6.1 SO 6	Ability to apply pseudocode	25%	The pseudocode written solves at least 90% of the programming problem	The pseudocode written solves at least 75% of the programming problem	The pseudocode written solves at least 60% of the programming problem	The pseudocode written solves less than 60% of the programming problem		
2	LO 3 LObj 6.1 SO 6	Ability to collect the requirements for a system	25%	All the necessary requirement of a system is collected	Some of the requirement of a system is collected	A few of the requirement of a system is collected	The requirement of a system is not collected		

No	Related LO- LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
4	LO 4 LObj 6.2 SO 6	Ability to design a system using UML	25%	Able to design at least 90% correct using UML	Able to design at least 75% correct using UML	Able to design at least 60% correct using UML	Able to design less than 60% correct using UML		
Total Score: $\sum(\text{Score} \times \text{Weight})$									

Remarks:

ASSESSMENT METHOD

Instructions:

- This project requires a group of three students to work together.
- After receiving a case study, students must::
 - Identify program requirements
 - Draw a UML diagram(s) to describe the process
 - Develop the algorithm using pseudocode.
- The assignment is compiled and presented in a singular .pdf format, subsequently being submitted through the binusmaya.

Note for Lecturers:

- It is advisable to inform the students to establish a group during the initial meeting in order to promptly commence their collaborative efforts on the assignment.
- In order to facilitate the submission of reports, lecturers have the ability to establish groups within the binusmaya platform.
- The submission of the report, ideally one week prior to the final exam, is to be carried out by a single student who will act as the representative for each group.

QUESTIONS

Case Study (100%)

EV Electric Motorcycle Battery Rental Application

Because the trend toward electric vehicles is expanding but has not been accompanied by the development of Public Electric Vehicle Charging Stations, many traditional vehicle users are still unwilling to transition to electric vehicles. A new company sees this problem as an opportunity for a rental business, which is also expected to provide vehicle battery pick-up and drop-off services. Initially, this company exclusively sells batteries for specific manufacturers of electric motorcycles. Battery types will gradually evolve to meet market demands. Users can visit the rental location or use the app to arrange for pick-up and drop-off. To use this service, consumers must first register and make a deposit, which serves as collateral in the event of loss or damage to the rented battery. Outlets also offer services for selling (new or used) or purchasing old batteries from customers, thereby reducing environmental damage caused by illicit battery disposal. In this assignment, it is recommended that you construct the same application but include unique features that distinguish your product from competitors' applications.

Do some of the things below as follows

1. **[LO 1 : 25 poin]** Identify the needs of the application mentioned above.
2. **[LO 4 : 25 poin]** Create a Use Case Diagram and an Activity Diagram for each use case.
3. **[LO 3 : 25 poin]** Create a Class Diagram (including attributes and methods) and provide pseudocode for the methods.
4. **[LO 2: 25 poin]** Describe the process you would use to gather requirements for the application, and include a state transition diagram representing the battery life cycle.

-- Good luck --

1. Identify the needs of the application mentioned above.

1. Registrasi Pengguna

- Pengguna perlu mendaftar melalui aplikasi lalu memberikan informasi dan untuk menyetujui ketentuan yang berlaku

2. Sistem Deposit

- Pengguna memasukkan uang sebagai deposit, yang dimana deposit tersebut merupakan tanda jaminan atas kehilangan atau kerusakan baterai sewa.
- Uang tersebut bisa ditransfer melalui qris yang akan ditampilkan setelah pengguna melakukan registrasi
- Sistem otomatis untuk mencatat, mengembalikan, atau memotong deposit sesuai kondisi baterai.

3. Layanan Penyewaan Baterai

- Pengguna bisa mendatangi lokasi rental yang ada atau menggunakan aplikasi untuk mengatur pick-up dan juga drop-off
- Selain menyewa baterai, pengguna bisa melakukan pembelian baterai baik baru maupun bekas dan juga membeli baterai bekas dari pengguna.
- Informasi lengkap tentang penggunaan baterai, termasuk tanggal sewa, pengembalian, dan kondisi baterai.\
- Pengguna bisa mengcancel jika tidak jadi menggunakan jasa penyewaan

4. Pengelolaan Baterai

- Sistem untuk mencatat stok baterai baru, bekas, atau baterai yang tersedia untuk dijual atau disewa.
- Sistem untuk melacak kesehatan dan kondisi baterai.

5. Informasi Lokasi dan Navigasi

- Fitur yang memungkinkan pengguna menemukan outlet terdekat.
- Navigasi untuk pick-up atau drop-off baterai.

6. Sistem Antarmuka Aplikasi

- Sistem yang mudah digunakan oleh pengguna dan navigasi yang mudah sehingga pengguna dapat

7. Sistem Notifikasi

- Notifikasi mengenai promo yang sedang berlaku, berapa lama telah menyewa, dan waktu untuk mengembalikan baterai.

8. Pengamanan Data Pengguna

- Sistem untuk mengamankan informasi pengguna, transaksi pengguna, pengelolaan akses, dan pencandangan data

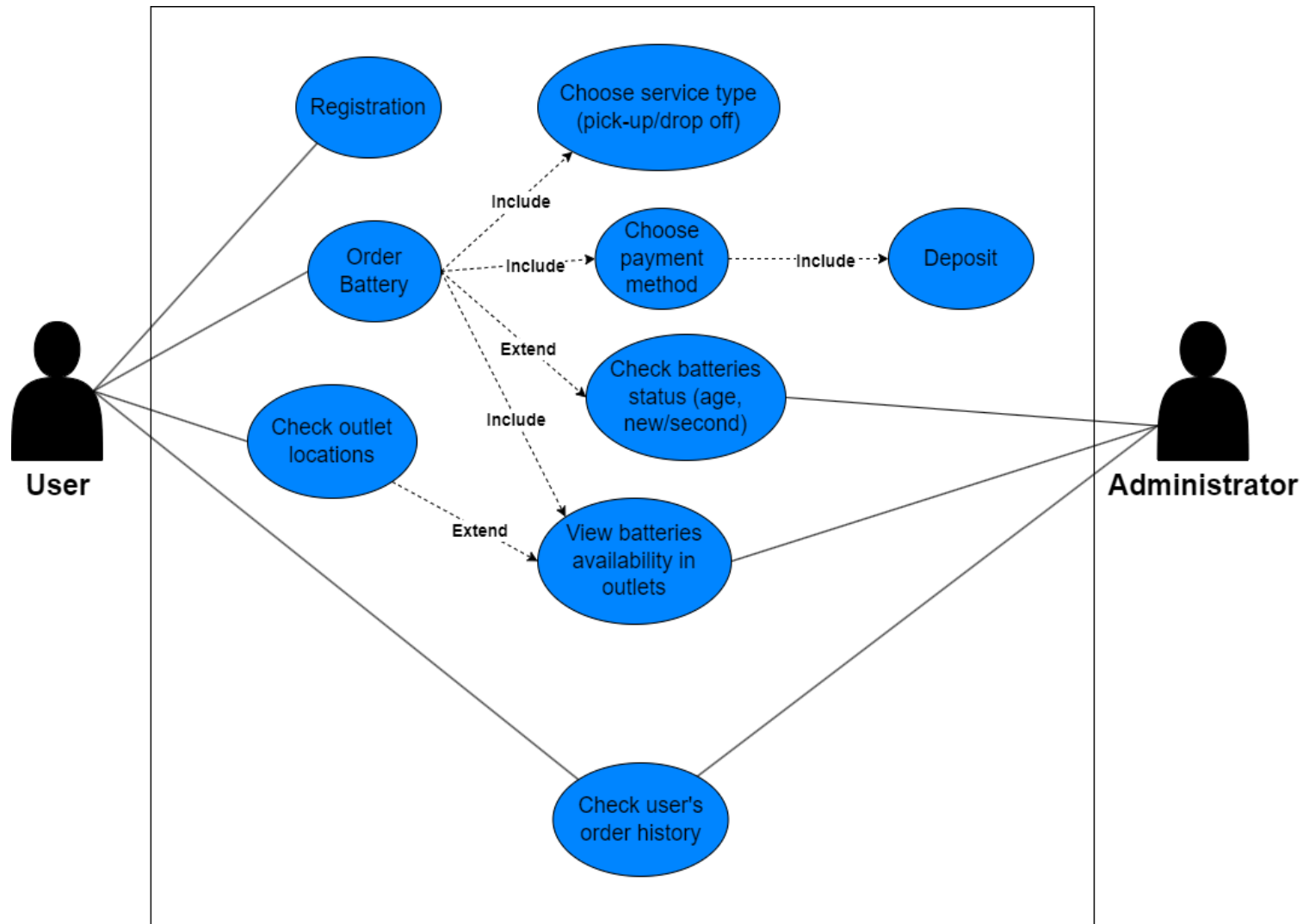
9. Sistem Pembayaran

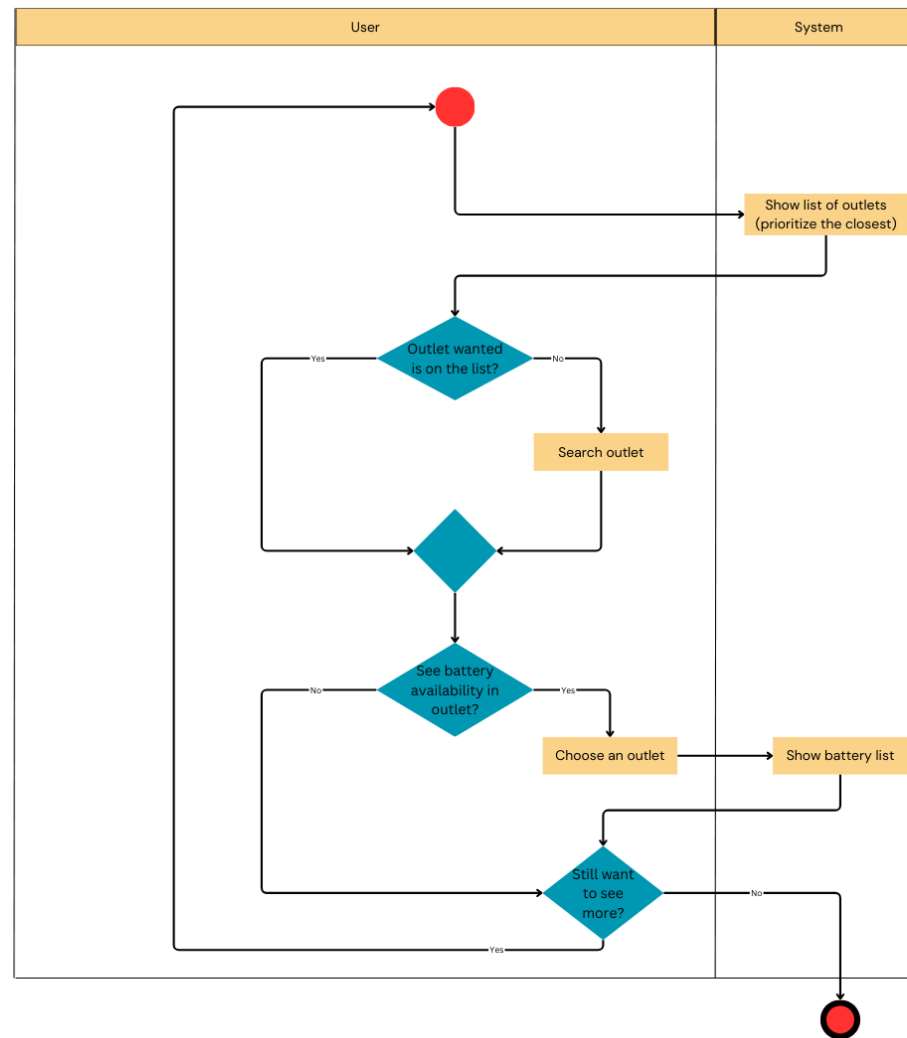
- Pembayaran dapat menggunakan e-wallet, transfer bank, QR, dan secara langsung.

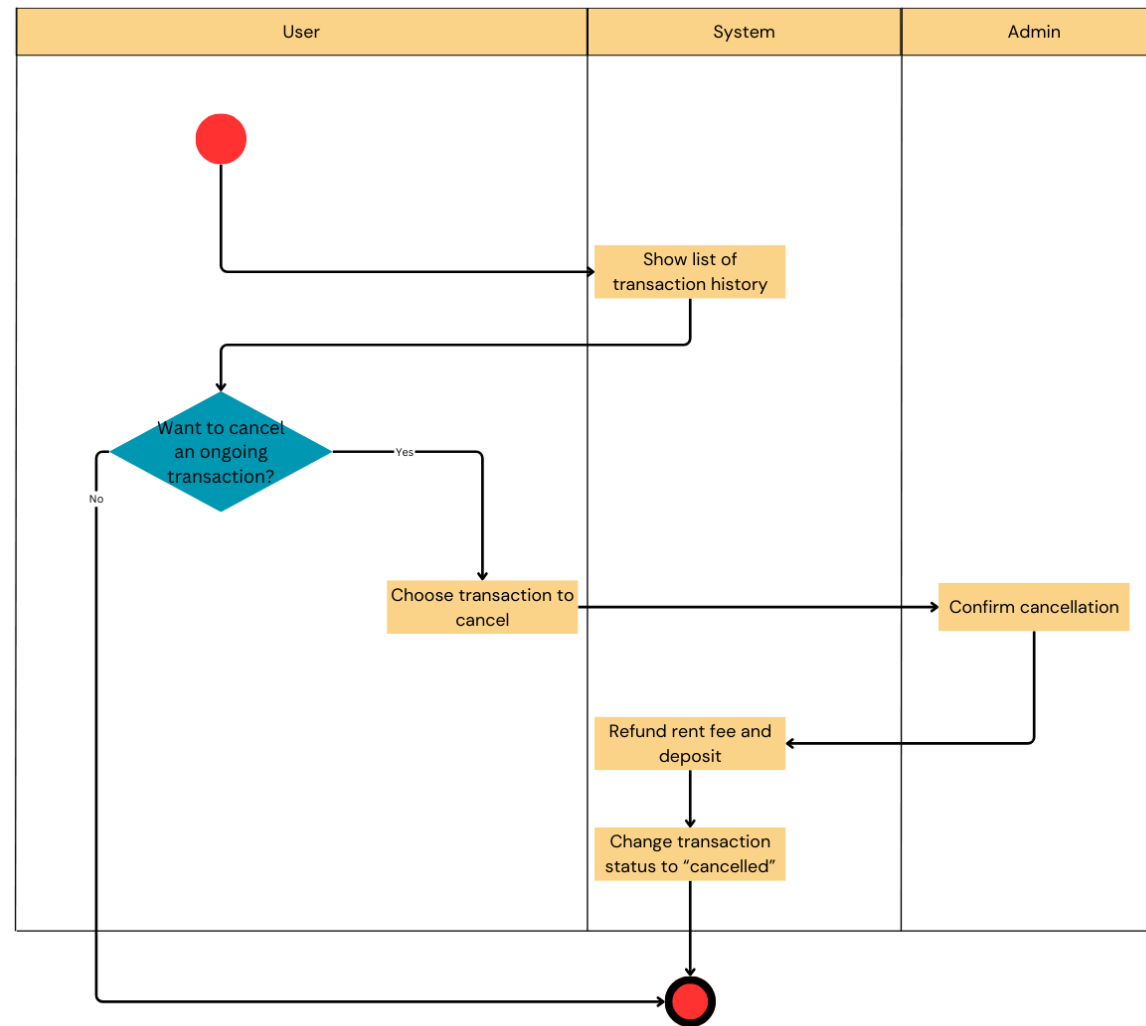
10. Fitur Unik (Distinctive Features)

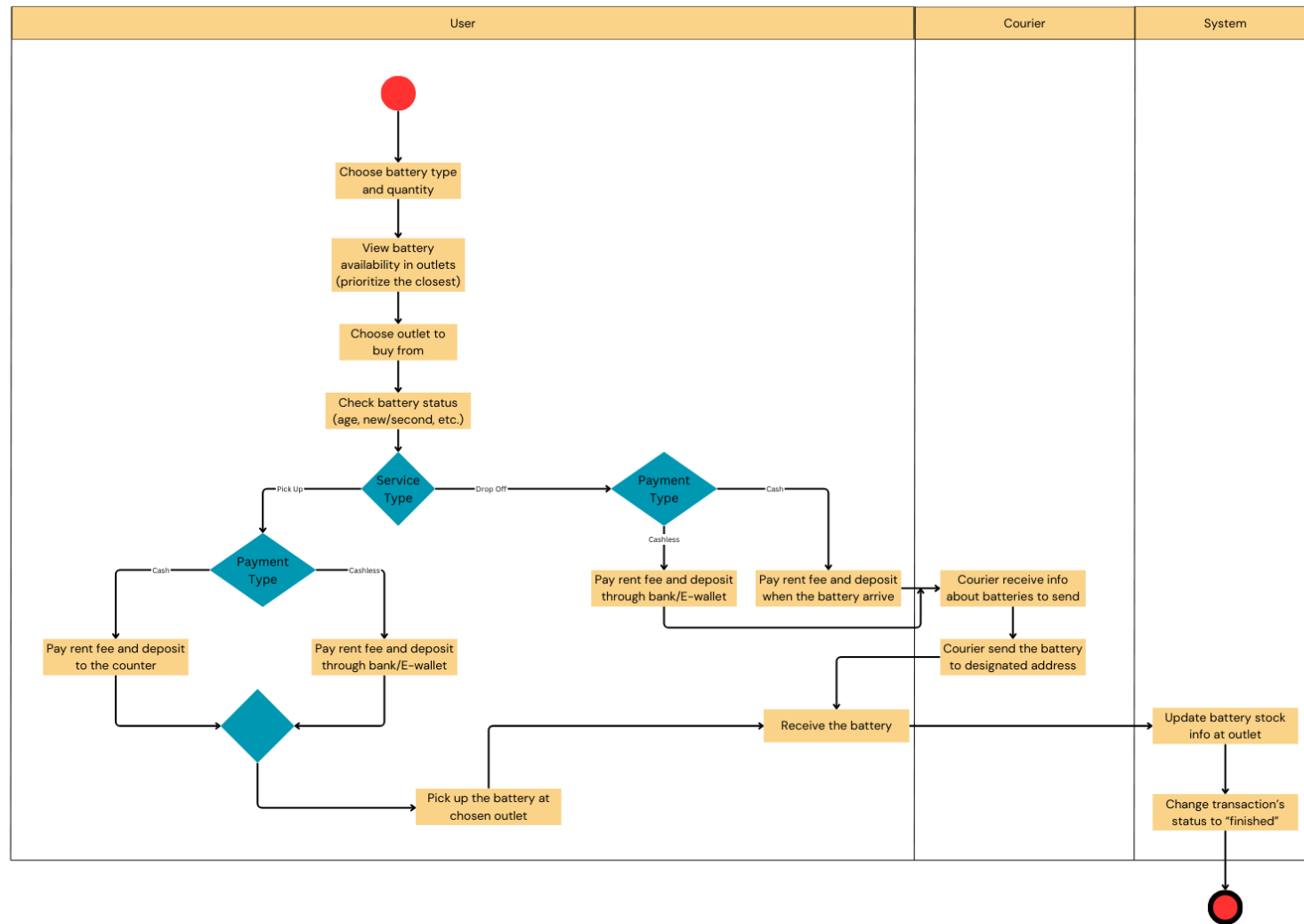
- Fitur langganan yang membuat pengguna dapat berlangganan untuk mendapatkan harga yang lebih murah dibanding tidak berlangganan. Pengguna diberikan sebuah akses video tutorial untuk tata cara menggunakan aplikasi, saat membuka aplikasi tersebut video tersebut akan muncul untuk memberi tau cara menggunakan aplikasi tersebut. Adanya sistem poin yang akan diberikan ketika telah menyelesaikan transaksi. Poin tersebut dapat dikumpulkan dan dapat digunakan pengguna untuk potongan diskon.

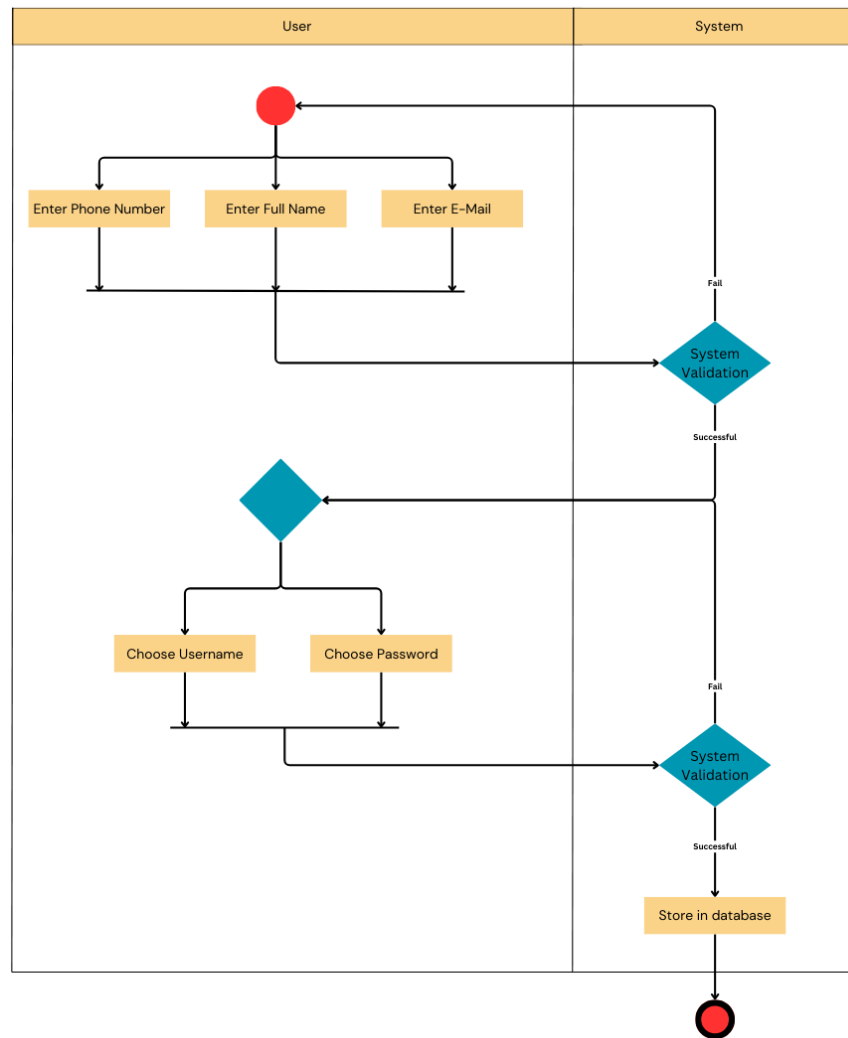
2. Use Case Diagram and an Activity Diagram for each use case.



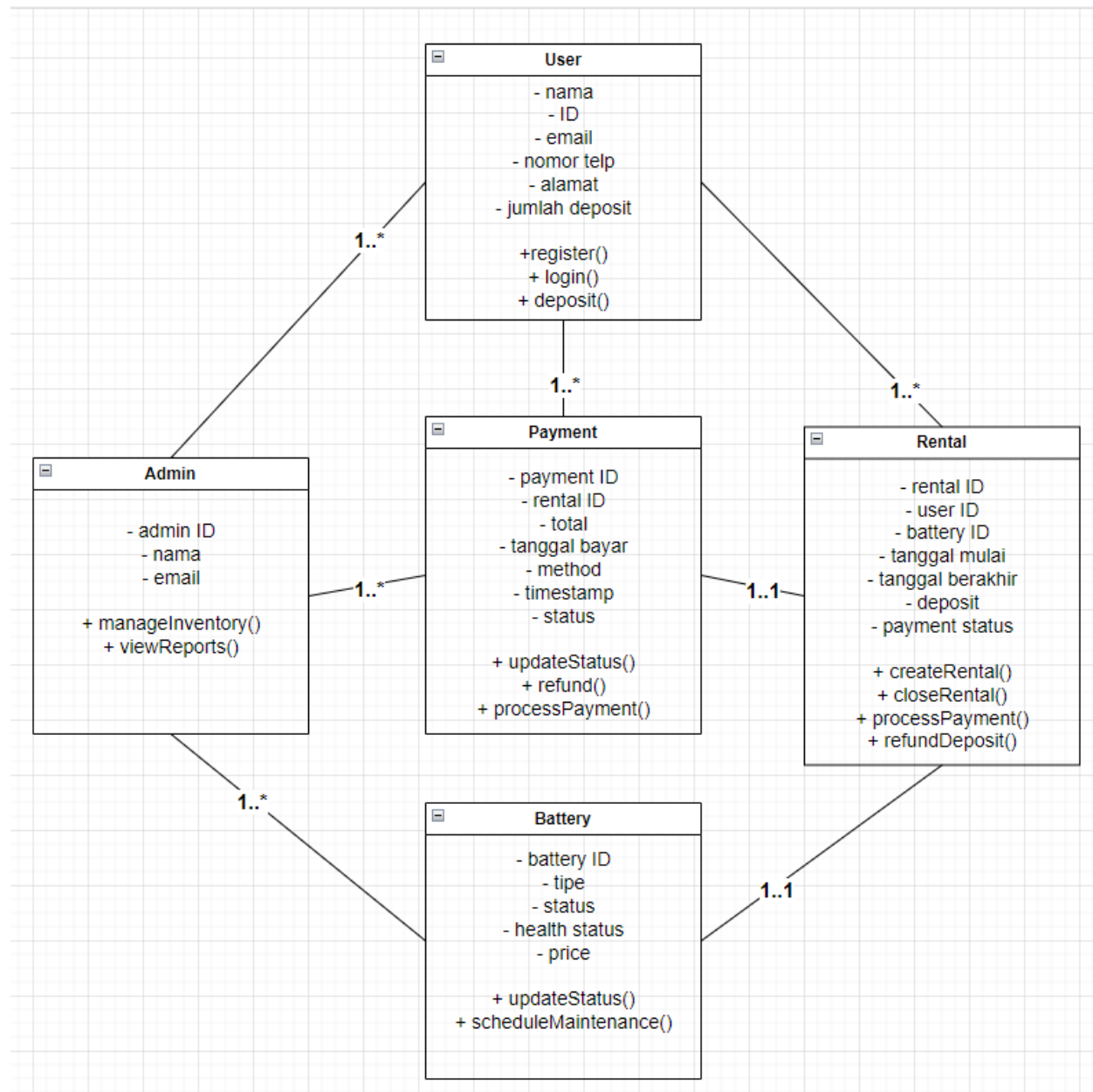








3. Class Diagram (including attributes and methods) and provide pseudocode for the methods.



```
# Customer Methods
Customer.register():
    Print "Customer registered successfully"

Customer.update_contact_info(new_contact_info):
    Set self.contact_info = new_contact_info
    Print "Contact information updated"

# Battery Methods
Battery.update_status(new_status):
    Set self.status = new_status
    Print "Battery status updated"

# Rental Methods
Rental.calculate_total_price():
    Set rental_days = (self.end_date - self.start_date).days
    Return rental_days * self.battery.price

# Outlet Methods
Outlet.add_battery(battery):
    Append battery to self.batteries
    Print "Battery added to outlet inventory"

Outlet.remove_battery(battery_id):
    Set self.batteries = [b for b in self.batteries if b.battery_id != battery_id]
    Print "Battery removed from outlet inventory"

# App Methods
App.register_customer(customer):
    Append customer to self.customers
    Print "Customer registered in the app"

App.find_nearest_outlet(location):
    # Logic to find nearest outlet based on location
    Print "Nearest outlet found"
    Return nearest_outlet

App.arrange_pickup_or_dropoff(customer_id, battery_id, action):
    # Logic to arrange pick-up or drop-off
    Print f"{action} arranged for customer {customer_id} and battery {battery_id}"
```

4. Describe the process you would use to gather requirements for the application, and include a state transition diagram representing the battery life cycle.

- 1. Melakukan Survey Konsumen**

- Target konsumen yang akan dituju
- Area yang akan dipasangkan tempat rental
- Aplikasi yang diinginkan konsumen
- Melihat kebutuhan konsumen

- 2. Melakukan riset**

- Melihat pesaing yang sudah ada
- Mencatat fitur – fitur yang penting yang dapat diimplementasikan kepada aplikasi sendiri
- Melihat tanggapan konsumen terhadap pesaing dan melihat saran – saran dari konsumen

- 3. Melihat Aturan yang berlaku**

- Kebijakan yang dilakukan pemerintah
- Apa saja yang boleh dan tidak boleh dilakukan
- Regulasi pengelolaan baterai

- 4. Membuat sebuah dokumentasi**

- Dokumentasi dalam yang menggambarkan fungsi aplikasi secara utuh
- Dokumentasi bisa dalam berbentuk diagram ataupun flowchart.

